

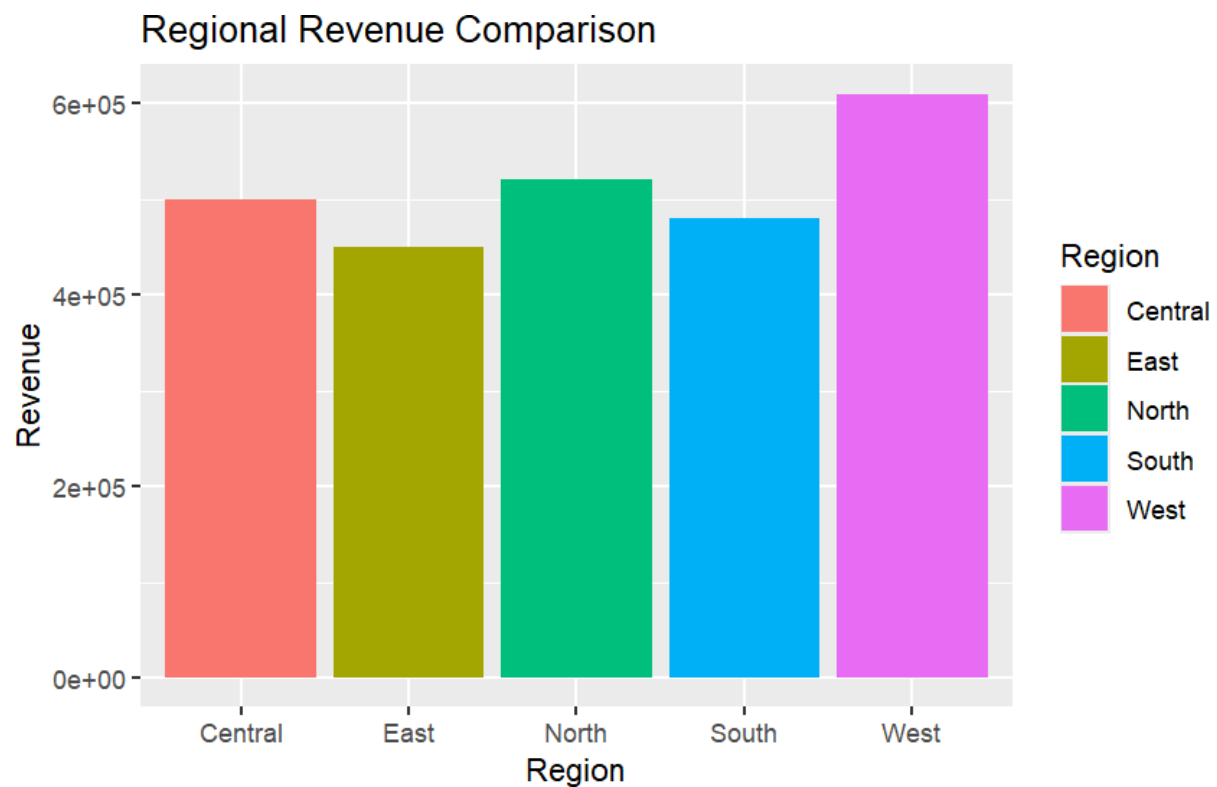
DSA0601-DATA HANDLING AND VISUALIZATION

CO2_AT3- CASE STUDY

CASE STUDY 1 – Retail Sales Performance Dashboard

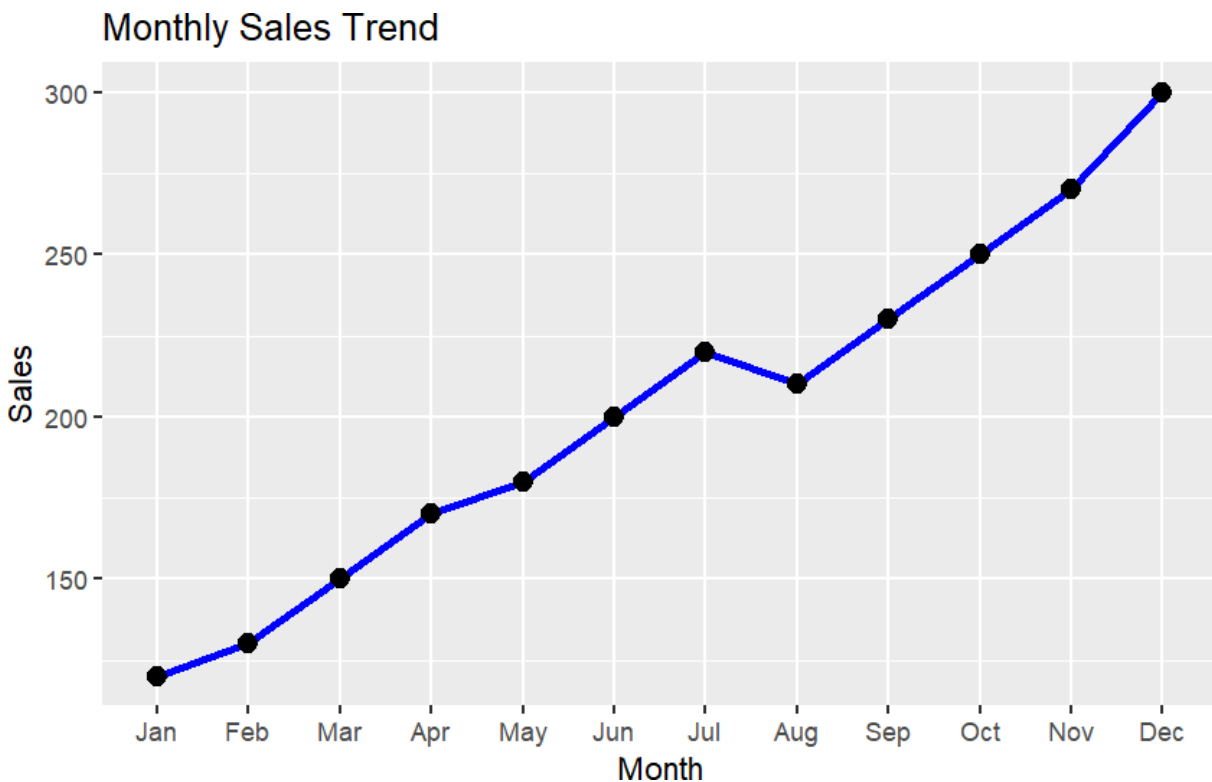
Requirement	Best Visualization
Regional revenue comparison	Bar Plot
Monthly sales trend	Line Chart
Product category contribution	Stacked Bar Chart
Advertising expenditure vs Sales	Scatter Plot
Regional sales distribution	Histogram

1. Regional Revenue Comparison – Bar Plot



CODE:

```
library(ggplot2)
sales <-
data.frame(
  Region =
    c("North","South","East","West","Central"),
  Revenue =
    c(520000,480000,450000,610000,500000)
)
ggplot(sales, aes(x=Region, y=Revenue,
  fill=Region)) + geom_bar(stat="identity")
+
labs(title="Regional Revenue Comparison",
  x="Region",
  y="Revenue")
```

2. Monthly Sales Trend – Line Chart**CODE:**

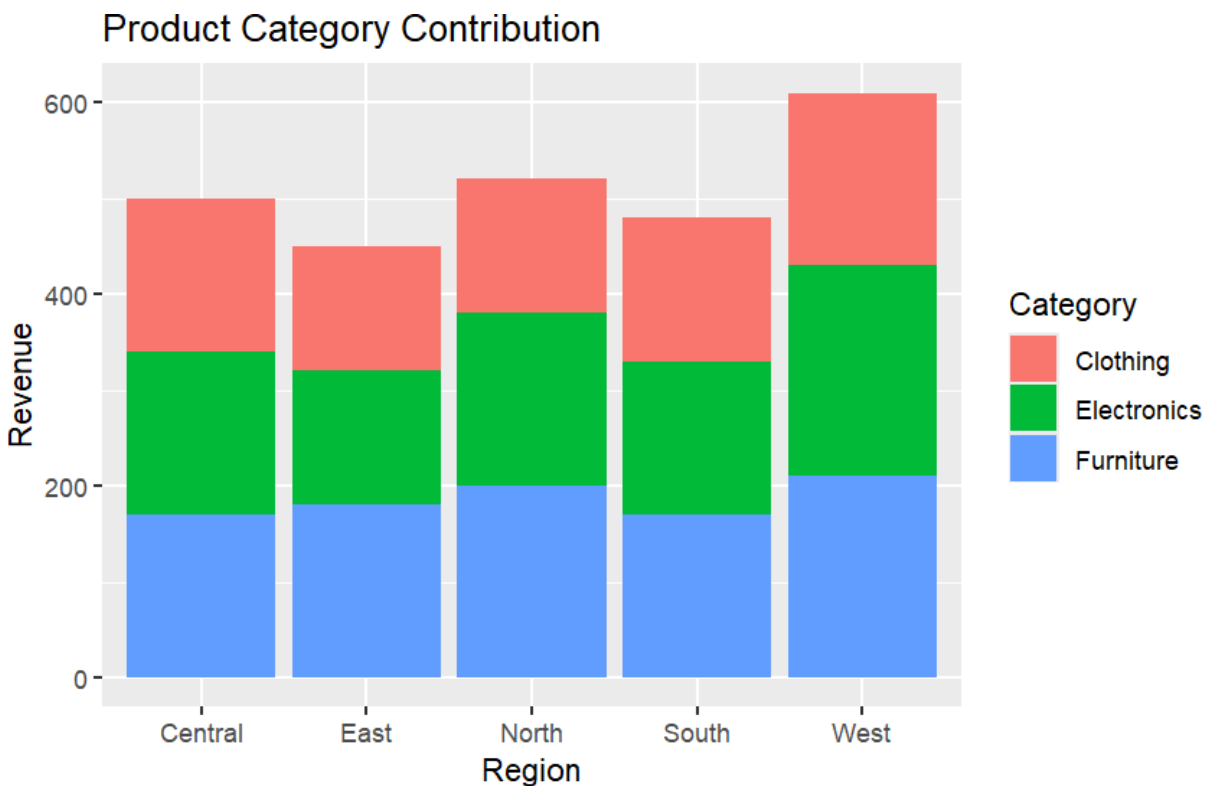
```
monthly <- data.frame(
```

```

Month = factor(month.abb, levels=month.abb),
Sales = c(120,130,150,170,180,200,220,210,230,250,270,300)
)
ggplot(monthly, aes(x=Month, y=Sales,
group=1)) + geom_line(color="blue",
linewidth=1.2) + geom_point(size=3) +
labs(title="Monthly Sales Trend",
x="Month",
y="Sales")

```

3. Product Category Contribution – Stacked Bar Chart



CODE:

```

category <- data.frame(
  Region=rep(c("North","South","East","West","Central"),
  each=3),
  Category=rep(c("Electronics","Clothing","Furniture"),5),
  Revenue=c(180,140,200,
    160,150,170,
    140,130,180,

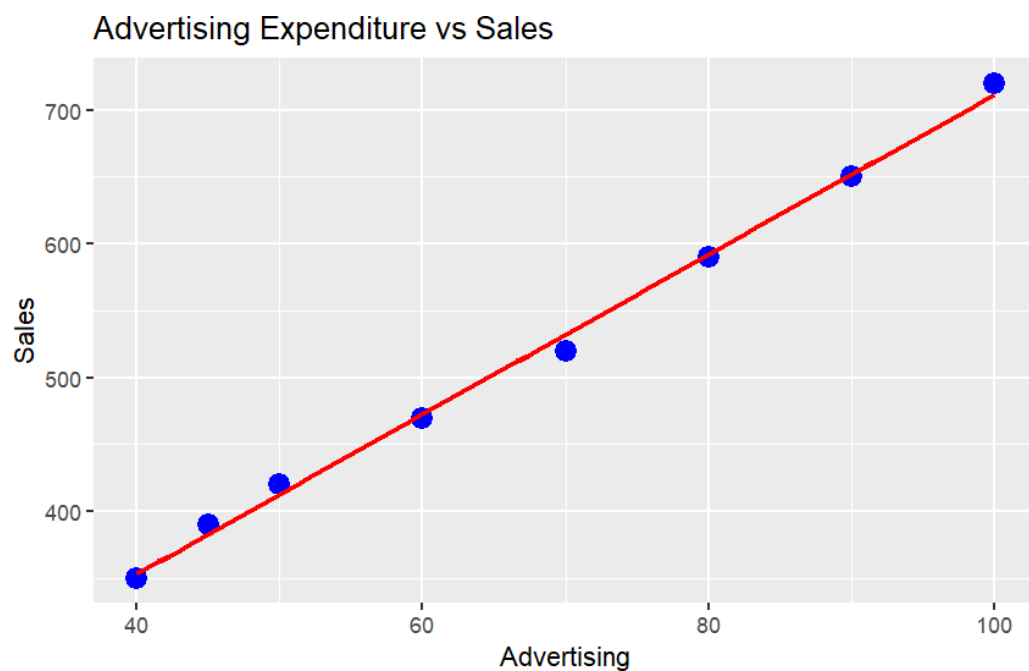
```

```

220,180,210,
170,160,170)
)
ggplot(category,
  aes(x=Region,y=Revenue,fill=Category))
+
geom_bar(stat="identity")+
labs(title="Product Category
Contribution")

```

4. Advertising Expenditure vs Sales – Scatter Plot



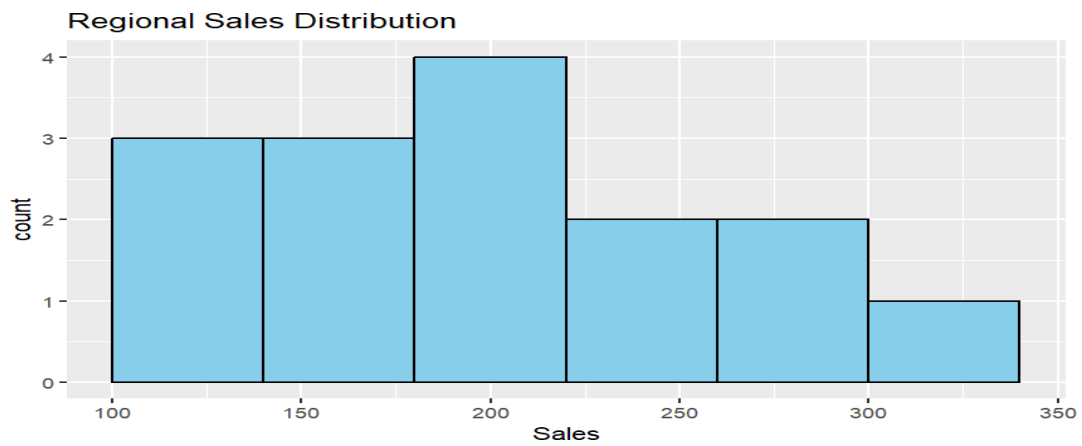
CODE:

```

ads <- data.frame(
  Advertising=c(40,45,50,60,70,80,90,100),
  Sales=c(350,390,420,470,520,590,650,720)
)
ggplot(ads,
  aes(x=Advertising,y=Sales))+
geom_point(size=4,color="blue")+
geom_smooth(method="lm",se=FALSE,col
or="red")+ labs(title="Advertising
Expenditure vs Sales")

```

5. Regional Sales Distribution – Histogram



CODE:

```
salesdist <- data.frame(  
  Sales=c(120,130,140,150,160,180,190,200,2  
    10,220,  
      240,260,280,300,320)  
)  
ggplot(salesdist,aes(Sales))+  
  geom_histogram(fill="skyblue",  
    color="black  
    ", bins=6)+  
  labs(title="Regional Sales Distribution")
```

1. Regional Revenue Comparison (Bar Plot)**Interpretation:**

- The West region generated the highest revenue.
- The East region recorded the lowest revenue.
- The company should analyze the success factors in the West region and apply similar strategies to lower-performing regions.

2. Monthly Sales Trend (Line Chart)**Interpretation:**

- Sales show a steady increase throughout the year.
- The highest sales are observed in December.
- The increasing trend indicates growing customer demand and improved business performance.

3. Product Category Contribution (Stacked Bar Chart)**Interpretation:**

- Electronics contributes the largest share of total revenue.
- Clothing and Furniture contribute comparatively less.
- Electronics should remain a key focus for marketing and inventory planning.

4. Advertising Expenditure vs Sales (Scatter Plot)

Interpretation:

- The scatter plot shows a positive relationship between advertising expenditure and sales.

- As advertising spending increases, sales also tend to increase.
- This suggests that advertising has a positive impact on revenue generation.

5. Regional Sales Distribution (Histogram)

Interpretation:

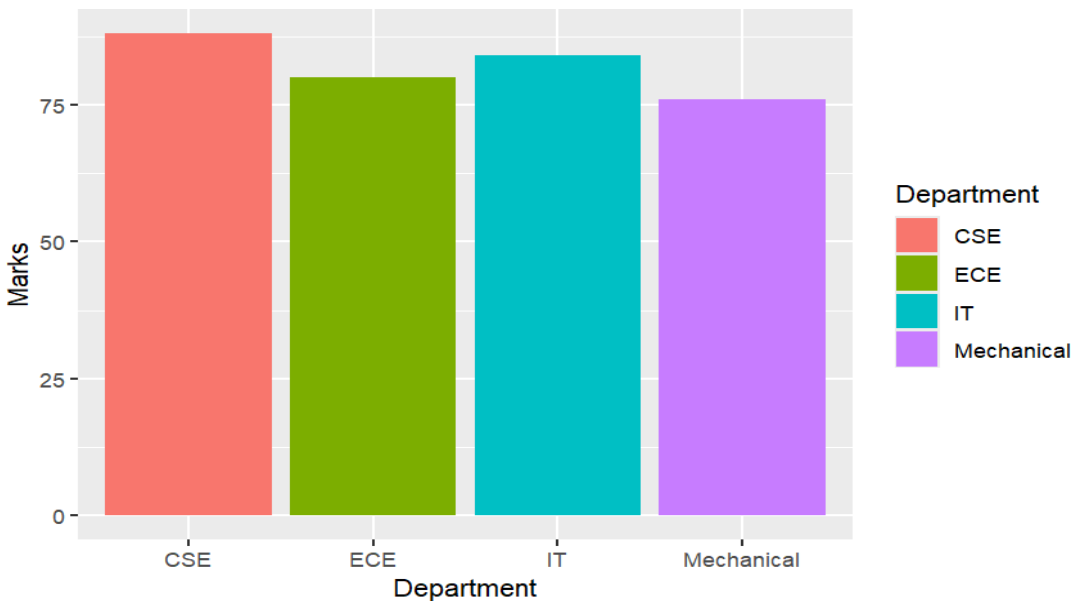
- Most sales values are concentrated in the medium revenue range.
- Very few regions have extremely high or extremely low sales

CASE STUDY 2 – University Student Performance

Question 1

Requirement	Best Visualization
Department-wise average marks	Grouped Bar Chart
Distribution of CGPA	Histogram
Placement percentage	Bar Chart
Attendance vs CGPA relationship	Scatter Plot
Semester-wise performance trend	Line Chart

1. Department-wise Average Marks



CODE:

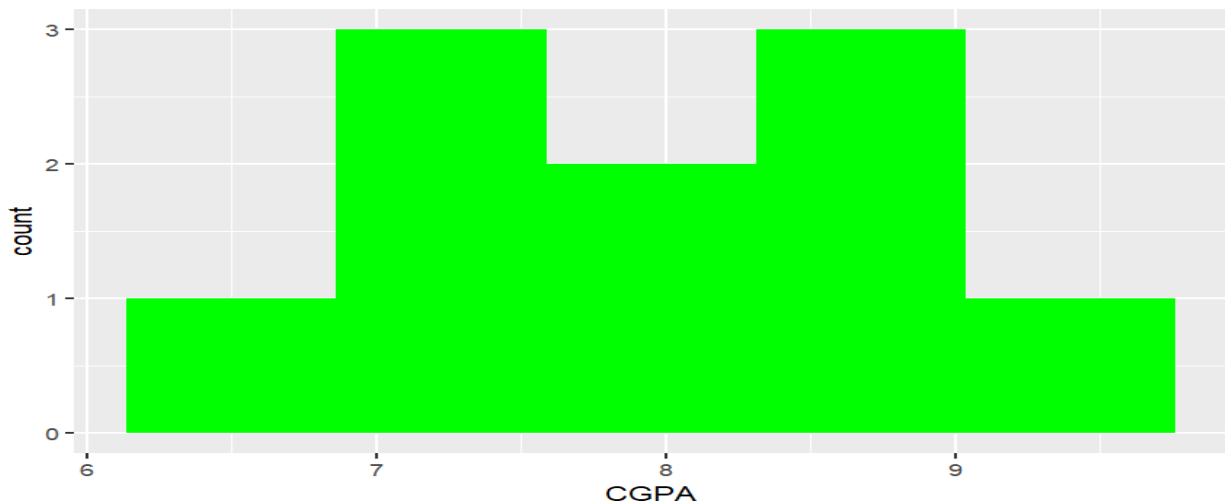
```
dept <- data.frame(
```

```
Department=c("CSE","IT","ECE","Mechanical"),
```

```
Marks=c(88,84,80,76))
```

```
ggplot(dept,aes(Department,Marks,fill=Department))+ geom_bar(stat="identity")
```

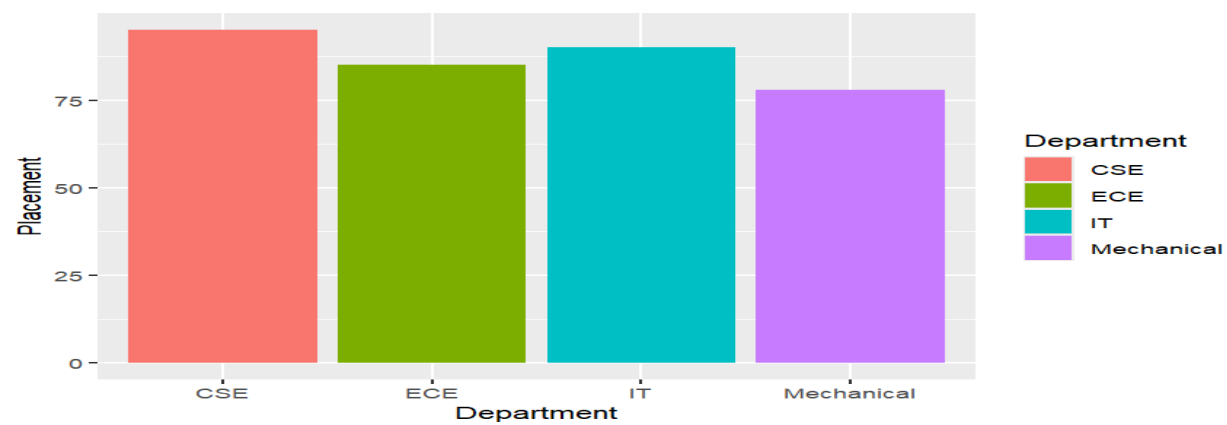
2. Distribution of CGPA



CODE:

```
cgpa <- data.frame(  
CGPA=c(6.5,7.0,7.2,7.5,8.0,8.2,8.5,8.8,9.0,9  
.4))  
ggplot(cgpa,aes(CGPA))+  
geom_histogram(fill="green",bins=5)
```

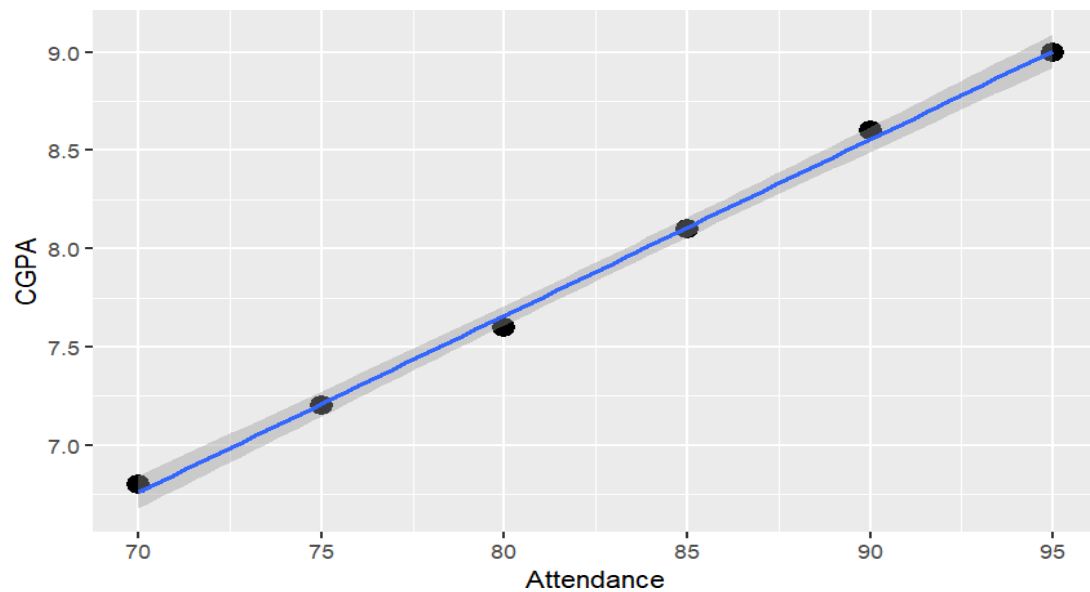
3. Placement Percentage



CODE:

```
placement <- data.frame(  
  Department=c("CSE","IT","ECE","Mechani  
cal"), Placement=c(95,90,85,78))  
  
ggplot(placement,  
  aes(Department,Placement,fill=Department))  
  
+ geom_bar(stat="identity")
```

4. Attendance vs CGPA



CODE:

```
student <- data.frame(  
  Attendance=c(70,75,80,85,90,95),  
  CGPA=c(6.8,7.2,7.6,8.1,8.6,9.0))  
  
ggplot(student,
```

```
aes(Attendance,CGP
```

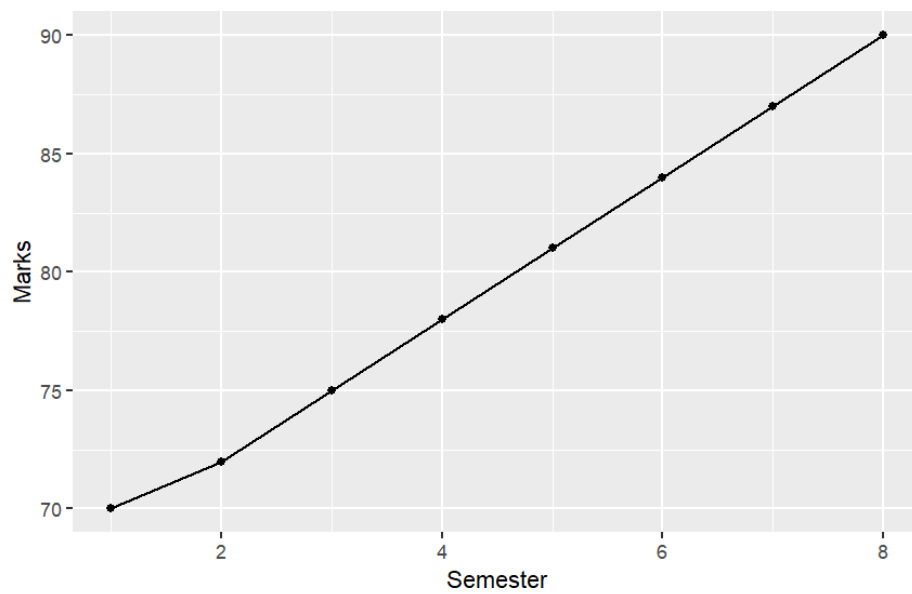
```
A))+
```

```
geom_point(size=4)
```

```
+
```

```
geom_smooth(method="lm")
```

5. Semester-wise Performance Trend



CODE:

```
semester <- data.frame(
```

```
Semester=1:8,
```

```
Marks=c(70,72,75,78,81,84,8
```

```
7,90))
```

```
ggplot(semester,
```

```
aes(Semester,Marks)
```

) + geom_line() +

geom_point()

1. Department-wise Average Marks (Grouped Bar Chart)

Interpretation:

- The **Computer Science** department has the highest average marks.
- The **Mechanical** department has the lowest average marks.
- Academic support can be provided to departments with lower performance.

2. Distribution of CGPA (Histogram)

Interpretation:

- Most students have a CGPA between **7.5 and 8.5**.
- Very few students have extremely low or extremely high CGPAs.
- The CGPA distribution indicates overall satisfactory academic performance.

3. Placement Percentage (Bar Chart)

Interpretation:

- The **Computer Science** department has the highest placement percentage.
- Mechanical has the lowest placement percentage.
- Placement training may be improved for departments with lower placement rates.

4. Attendance vs CGPA (Scatter Plot)

Interpretation:

- There is a **positive correlation** between attendance and CGPA.
- Students with higher attendance generally achieve better academic performance.
- Regular attendance contributes to improved learning outcomes.

5. Semester-wise Performance Trend (Line Chart)

Interpretation:

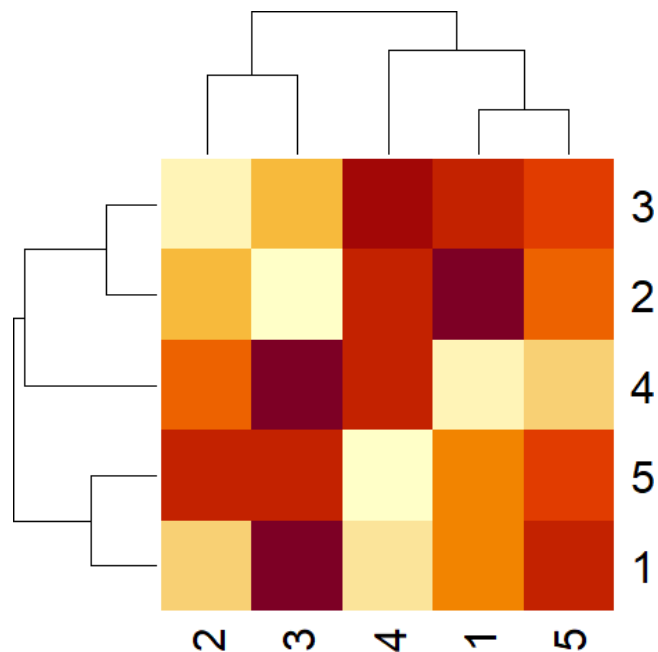
- Student performance improves gradually across semesters.
- Marks increase consistently from Semester 1 to Semester 8.
- This suggests continuous academic improvement over time.

CASE STUDY 3 – Hospital Patient Analytics

Question 1

Requirement	Best Visualization
Monthly patient admissions	Heatmap
Distribution of treatment costs	Histogram
Disease category proportions	Pie Chart
Age vs recovery time	Scatter Plot
Department-wise treatment costs	Violin Plot

Monthly Admissions (Heatmap)



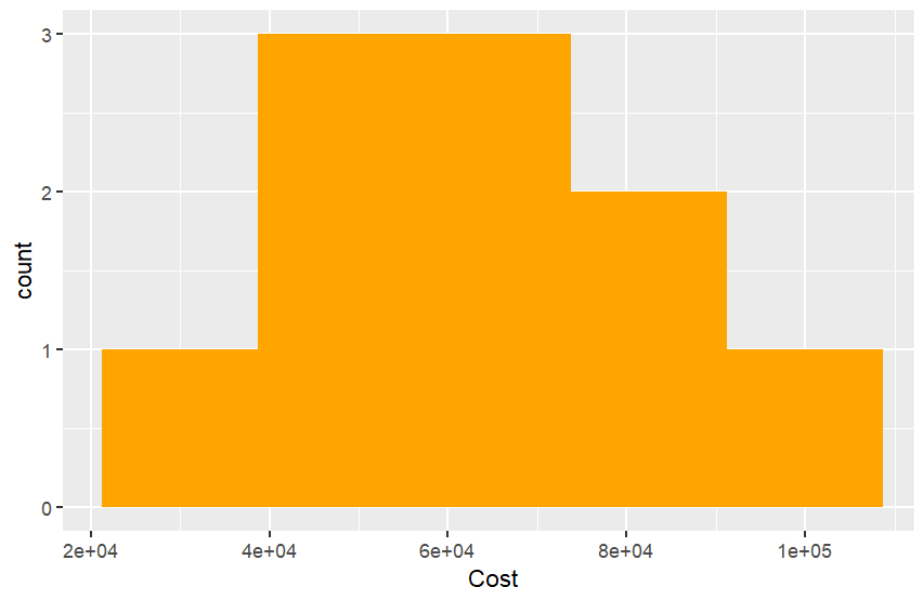
CODE:

```
data <- matrix(sample(10:100,25),nrow=5)
```



```
heatmap(data)
```

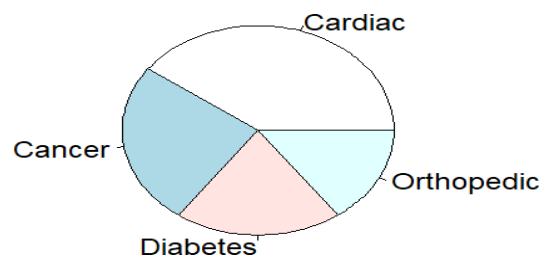
Treatment Cost Distribution



CODE:

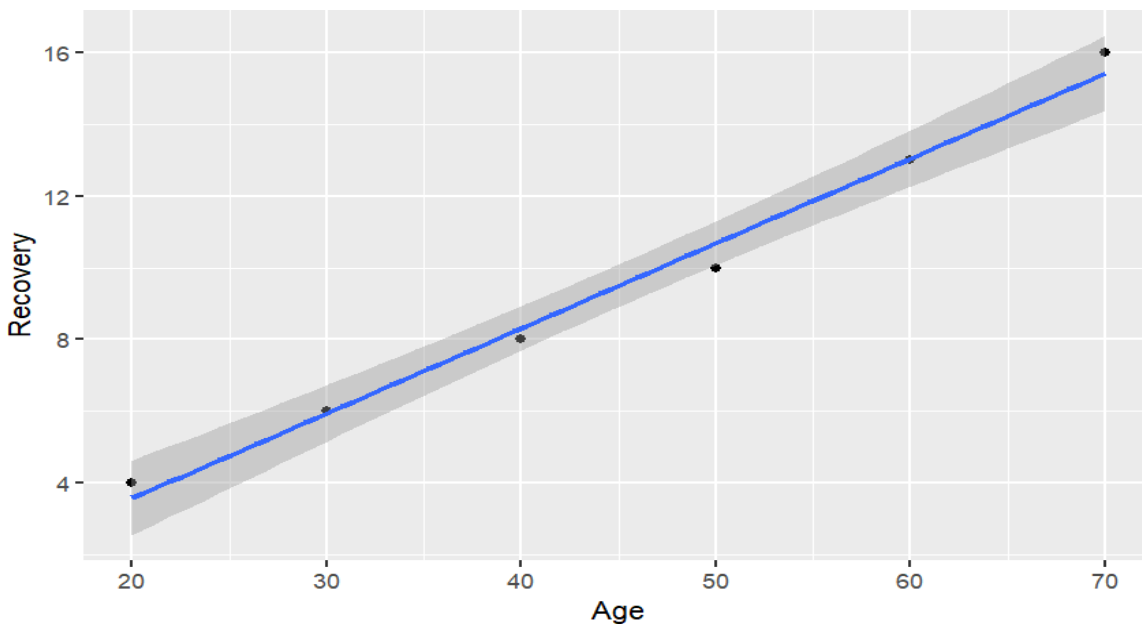
```
cost <- data.frame(  
  Cost=c(30000,40000,45000,50000,60000,  
  65000,70000,80000,90000,100000)  
)  
ggplot(cost,aes(Cost))+  
  geom_histogram(fill="orange",bins=5)
```

Disease Category Proportion



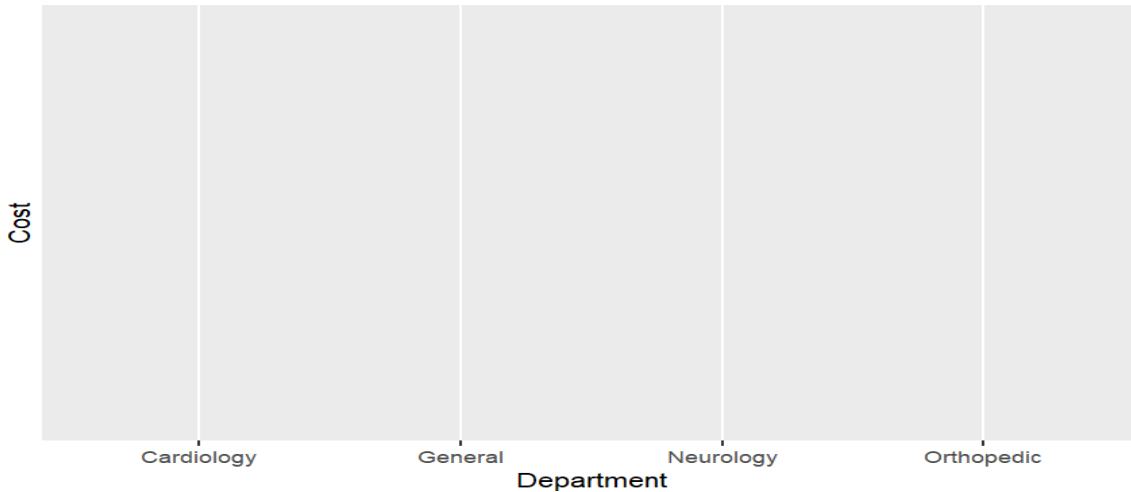
CODE:

```
disease <- c(40,25,20,15)
labels <-
c("Cardiac","Cancer","Diabetes","Orthopedi
c") pie(disease,labels=labels)
```

Age vs Recovery Time**CODE:**

```
patient <- data.frame(
  Age=c(20,30,40,50,60,70),
  Recovery=c(4,6,8,10,13,16))
ggplot(patient,
  aes(Age,Recovery))+
  geom_point()+
  geom_smooth(method="lm")
```

Department-wise Treatment Costs



CODE:

```
hospital <- data.frame(
  Department=c("Cardiology","Neurology",
    "Orthopedic","General"),
  Cost=c(90000,120000,75000,50000))
ggplot(hospital,
  aes(Department,Cost,fill=Departme
nt))+ geom_violin()
```

1. Monthly Patient Admissions (Heatmap)

Interpretation:

- Patient admissions vary across different months.
- Certain months experience higher admission rates than others.
- Hospital resources should be increased during peak admission periods.

2. Distribution of Treatment Costs (Histogram)

Interpretation:

- Most treatment costs fall within the moderate price range.
- Only a few patients incur very high treatment costs.
- The cost distribution helps estimate hospital expenses and budgeting.

3. Disease Category Proportions (Pie Chart)

Interpretation:

- **Cardiac diseases** represent the largest proportion of patients.
- Diabetes, Cancer, and Orthopedic cases make up smaller proportions.
- More medical resources should be allocated to the most common disease category.

4. Age vs Recovery Time (Scatter Plot)

Interpretation:

- Recovery time generally increases with patient age.
- Older patients tend to require longer recovery periods.
- Patient age should be considered when planning treatment and post-care.

5. Department-wise Treatment Costs (Violin Plot)

Interpretation:

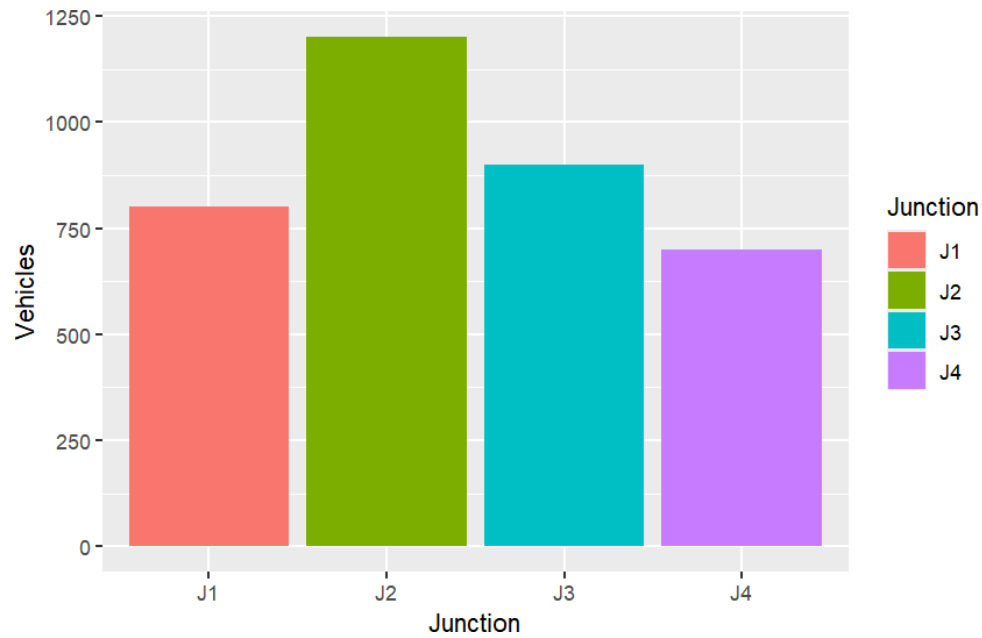
- **Neurology** has the highest treatment costs.
- General Medicine has the lowest treatment costs.
- High-cost departments may require more efficient resource management.

CASE STUDY 4 – Smart City Traffic Monitoring

Question 1

Requirement	Best Visualization
Vehicle count comparison	Bar Plot
Hourly traffic trend	Line Chart
Vehicle type proportion	Pie Chart
Traffic density by location	Geospatial Map
Traffic volume vs average speed	Scatter Plot

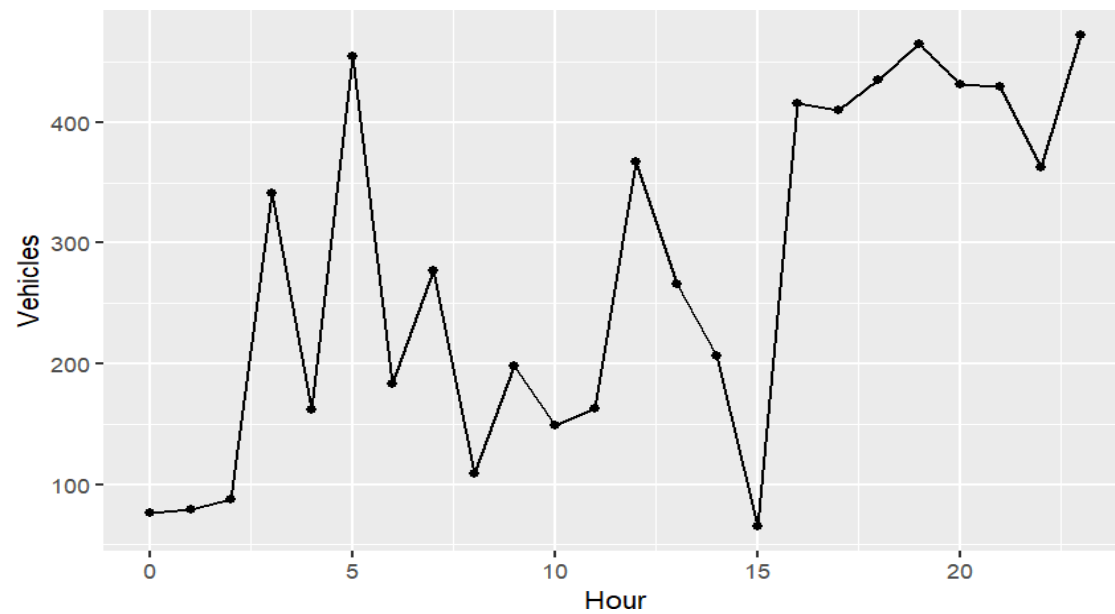
Vehicle Count Comparison



CODE:

```
traffic <- data.frame(
  Junction=c("J1","J2","J3","J4"),
  Vehicles=c(800,1200,900,700))
ggplot(traffic,
  aes(Junction,Vehicles,fill=Junction))
+
geom_bar(stat="identity")
```

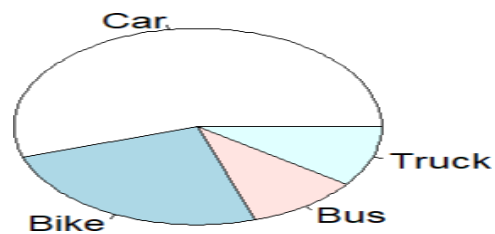
Hourly Traffic Trend



CODE:

```
hour <- data.frame(  
  Hour=0:23,  
  Vehicles=sample(50:50  
0,24)) ggplot(hour,  
aes(Hour,Vehicles))+  
geom_line()+  
geom_point()
```

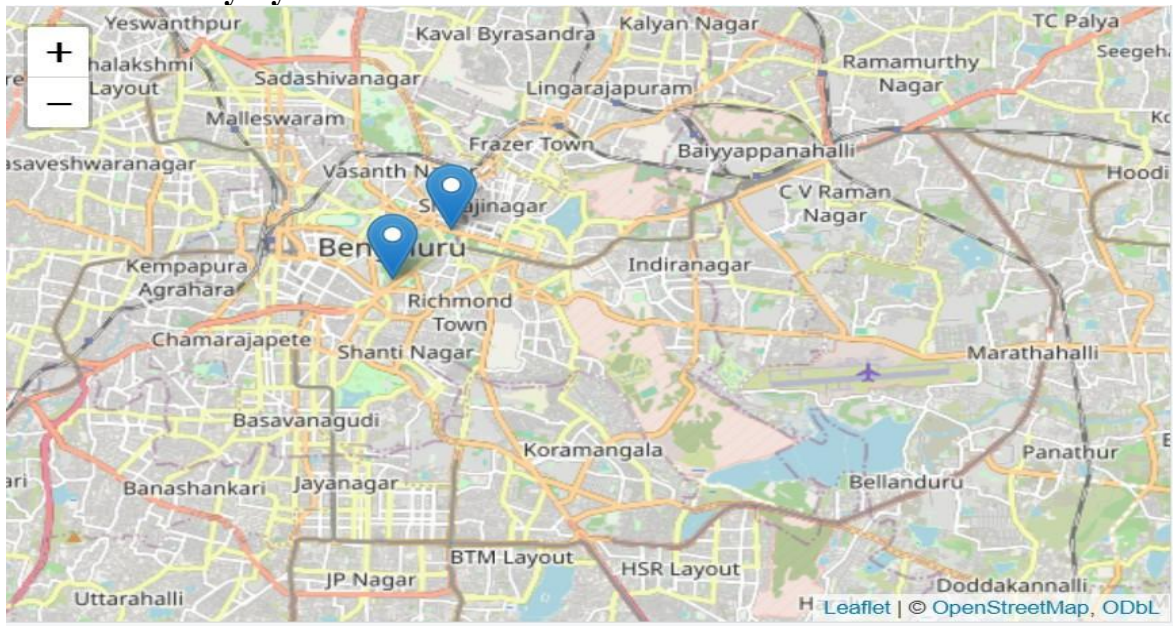
Vehicle Type Proportion



CODE:

```
vehicle <- c(55,25,10,10)  
labels <- c("Car","Bike","Bus","Truck")  
pie(vehicle,labels)
```

Traffic Density by Location



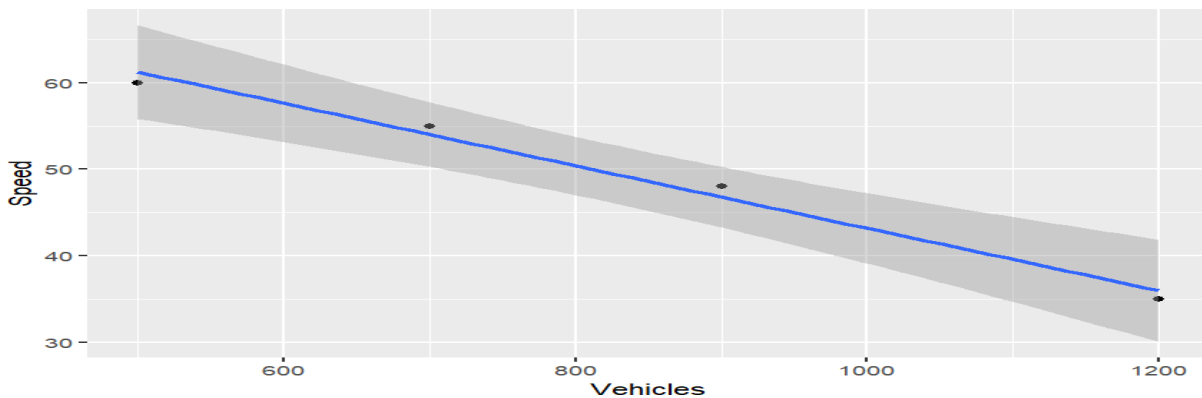
CODE:

```

library(leaflet)
leaflet() %>%
addTiles()
%>%
addMarkers(
  lng=c(77.59,77.
60),
  lat=c(12.97,12.98),
  popup=c("J1","J2")
)

```

Traffic Volume vs Speed



CODE:

```

speed <- data.frame(
  Vehicles=c(500,700,900,1200),
  Speed=c(60,55,48,35))
ggplot(speed,
  aes(Vehicles,Speed)) +
  geom_point() +
  geom_smooth(method="lm")

```

1. Vehicle Count Comparison (Bar Plot)

Interpretation:

- **Junction J2** has the highest vehicle count.
- Junction J4 has the lowest traffic volume.
- Traffic management efforts should prioritize J2.

2. Hourly Traffic Trend (Line Chart)

Interpretation:

- Traffic volume increases during the morning and evening peak hours.
- The highest traffic is observed between **5 PM and 7 PM**.
- Peak-hour traffic management measures can reduce congestion.

3. Vehicle Type Proportion (Pie Chart)

Interpretation:

- Cars make up the largest proportion of vehicles.
- Buses and trucks account for a much smaller share.
- Road planning should primarily consider the high volume of cars.

4. Traffic Density by Location (Geospatial Map)

Interpretation:

- Some junctions experience higher traffic density than others.
- Congestion hotspots can be easily identified using the map.
- Traffic signals and road improvements can be focused on these locations.

5. Traffic Volume vs Average Speed (Scatter Plot)

Interpretation:

- There is a **negative relationship** between traffic volume and average speed.
- As traffic volume increases, average vehicle speed decreases.
- High congestion significantly affects travel speed.

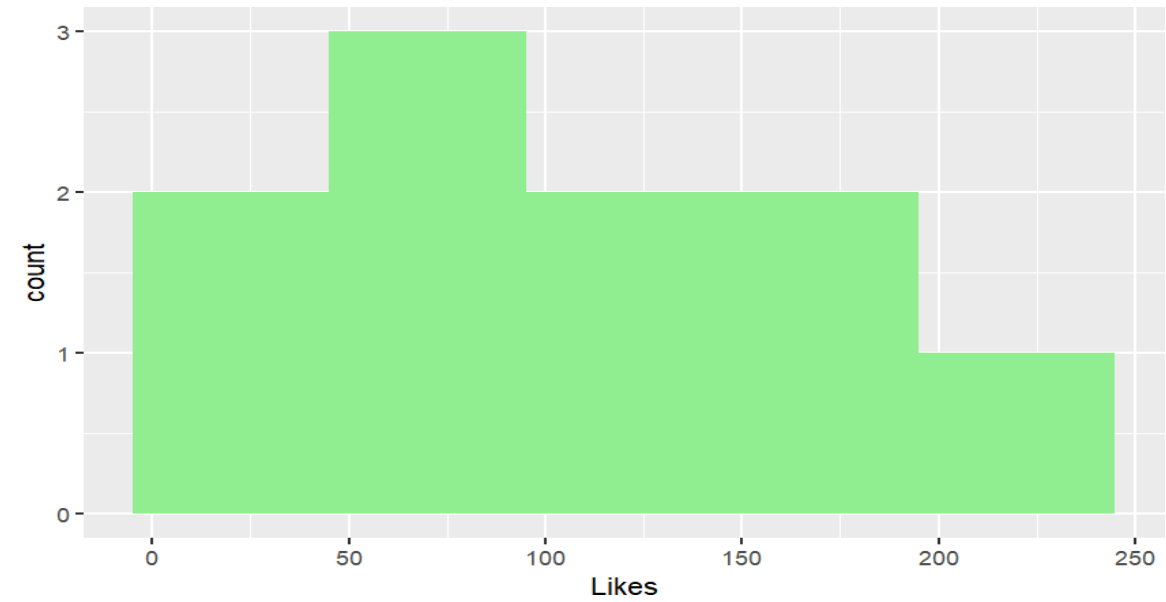
CASE STUDY 5 – Social Media User Interaction Analysis

Question 1

Requirement	Best Visualization
Distribution of likes	Histogram
Followers vs engagement	Scatter Plot

User interaction network	Network Graph
Posting activity over time	Line Plot
Likes, shares & comments comparison	Heatmap

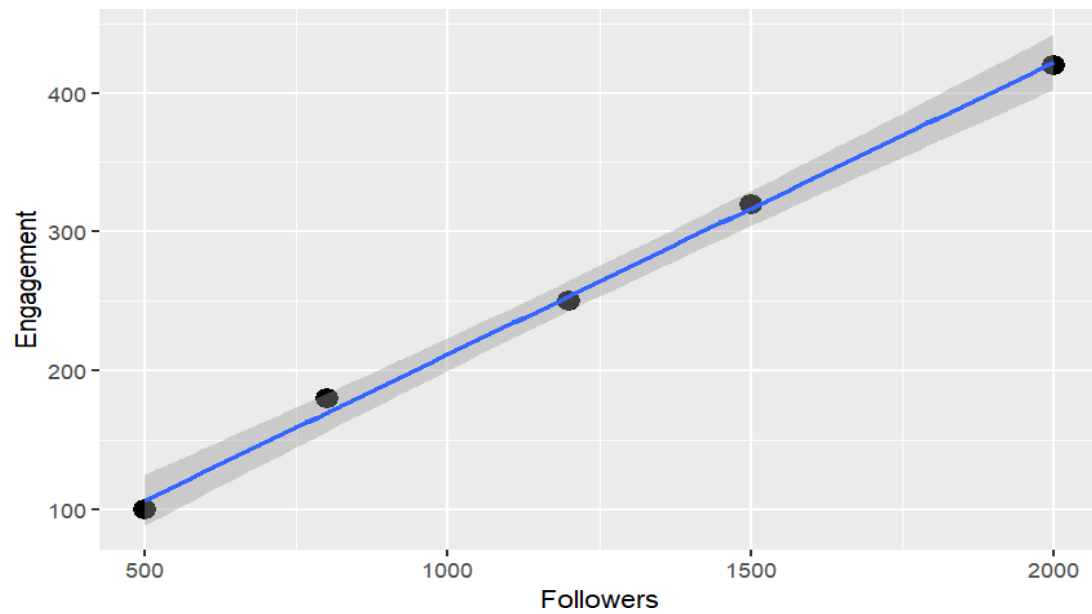
Distribution of Likes



CODE:

```
likes <- data.frame(  
Likes=c(20,30,50,70,80,100,120,150,180,220))  
ggplot(likes,aes(Likes))+  
geom_histogram(fill="lightgreen",bins=5)
```

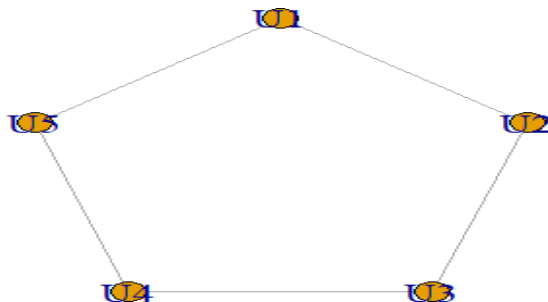
Followers vs Engagement



CODE:

```
social <- data.frame(
  Followers=c(500,800,1200,1500,2000),
  Engagement=c(100,180,250,320,420))
ggplot(social,
  aes(Followers,Engagement))
+ geom_point(size=4)+
  geom_smooth(method="lm")
```

User Interaction Network



CODE:

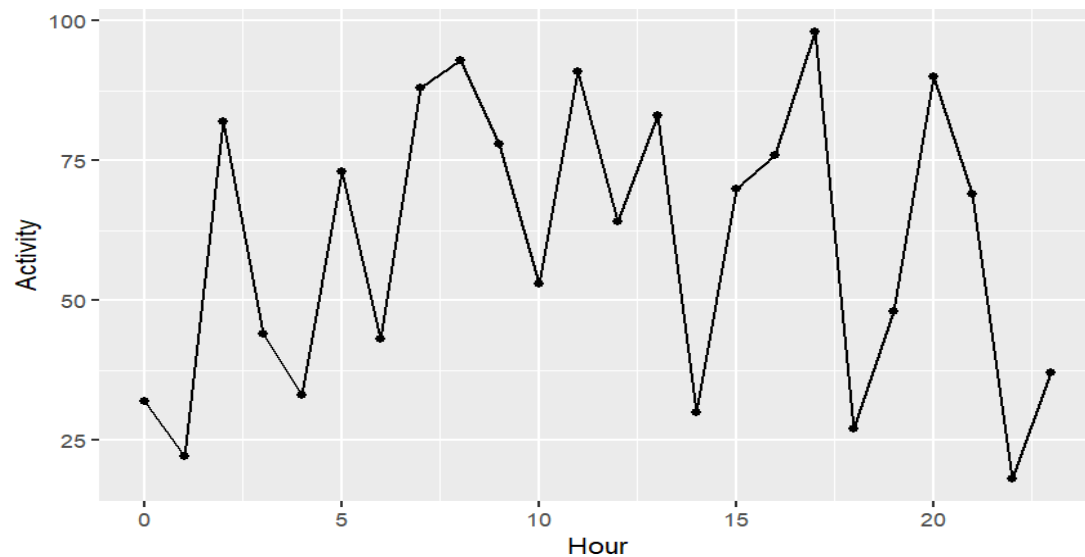
```
library(igraph)
g <- graph(edges=c(
  "U1","U2",
```

```

"U2","U3",
"U3","U4",
"U4","U5",
"U1","U5"
),
directed=FALS
E) plot(g)

```

Posting Activity Over Time



CODE:

```

posts <- data.frame(
  Hour=0:23,
  Activity=sample(10:100
,24)) ggplot(posts,
aes(Hour,Activity))+
geom_line()+
geom_point()

```

Likes, Shares and Comments

Comparison CODE:

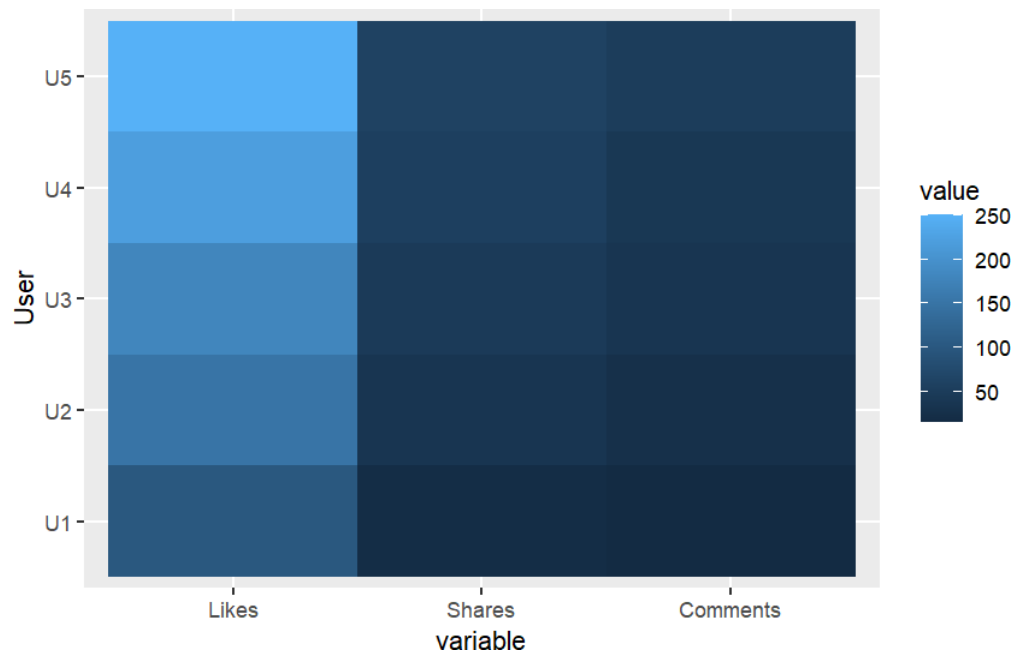
```

library(reshape2)
engagement <-
data.frame(
  User=c("U1","U2","U3","U4","U5")
, Likes=c(100,150,180,220,250),

```

```
Shares=c(20,35,45,55,60),
Comments=c(15,25,35,40,50))
melted <- melt(engagement,id="User")
```

```
ggplot(melted,
aes(variable,User,fill=value))
+ geom_tile()
```



1. Distribution of Likes (Histogram)

Interpretation:

- Most posts receive a moderate number of likes.
- Only a few posts achieve exceptionally high engagement.
- The distribution indicates that highly popular posts are relatively rare.

2. Followers vs Engagement (Scatter Plot)

Interpretation:

- There is a strong positive relationship between followers and engagement.
- Users with more followers generally receive higher engagement.
- Increasing follower count can improve overall user interaction.

3. User Interaction Network (Network Graph)

Interpretation:

- Some users have significantly more connections than others.
- These highly connected users are influential within the network.
- They play an important role in spreading information and increasing engagement.

4. Posting Activity Over Time (Line Plot)**Interpretation:**

- User activity changes throughout the day.
- Engagement reaches its peak during the evening hours (around 6 PM to 8 PM).
- Scheduling posts during these hours can improve audience interaction.

5. Likes, Shares, and Comments Comparison (Heatmap)**Interpretation:**

- Likes show the highest variation among users.
- Shares and comments are comparatively more consistent.
- The heatmap helps identify users with exceptionally high or low engagement across multiple metrics.